



Bahir Dar University
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Final Project Titles

- 1. Ketero-Tracker: A Real-Time Cloud-Based Queue & Appointment Management System for Small and Medium Enterprises**
- 2. Event-Sync: A Real-Time Multi-Vendor Event Coordination and Management Website**
- 3. ResearchSphere: A Research Management and Publication Tracking System for Higher Education Institutions**

Group Members

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Project Title 1: Ketero-Tracker: A Real-Time Cloud-Based Queue & Appointment Management System for Small and Medium Enterprises.

I. Introduction

In the fast-growing urban centers of Ethiopia, small and medium enterprises (SMEs) like barbershops, beauty salons, and boutique clinics are the backbone of the economy. However, these businesses struggle with "physical congestion" and "unstructured waiting." Ketero-Tracker is a lightweight, real-time web application built with the Firebase Realtime Database. It allows business owners to digitize their walk-in and scheduled appointments, providing customers with a live "Virtual Queue" accessible via a simple URL or QR code.

II. Statement of the Problem

Traditional service delivery in Ethiopian small shops is plagued by the "Wait-in-Person" culture, leading to several inefficiencies:

- ★ **Customer Time Poverty:** Clients are forced to sit in a shop for hours to "keep their spot," wasting productive time.
- ★ **Operational Overcrowding:** Physical waiting areas become congested, leading to poor ventilation and a stressful environment for both staff and customers.
- ★ **Scheduling Friction:** Relying on paper notebooks leads to double-bookings and makes it impossible to give customers an accurate "Estimated Time of Arrival" (ETA).
- ★ **Walk-away Loss:** Potential customers often leave when they see a crowded shop, resulting in lost revenue for the business owner.

III. Objectives

General Objective: To develop a real-time digital queue management system that eliminates physical waiting times for small business customers.

Specific Objectives:

To implement Instant Syncing using Firebase Realtime Database so that when a barber clicks "Next," all customers' screens update immediately.

To create a Dynamic ETA Algorithm that estimates wait times based on the average duration of the services in the queue.

To build a No-App-Required Interface (PWA) that allows customers to join a queue simply by scanning a QR code at the shop entrance.

To design a Merchant Dashboard where shop owners can manage their daily "Hisab" (records) and queue status with a single click.

IV. Significance of the Project

For Shop Owners, it increases efficiency and professionalizes their brand. For Customers, it provides "Time Freedom"—allowing them to run other errands or stay at home until their turn is near. Technically, this project demonstrates the use of Serverless Architecture and WebSockets to solve a high-traffic local problem with zero infrastructure cost.

Project Title 2: Event-Sync: A Real-Time Multi-Vendor Event Coordination and Management Website

I. Introduction

In the modern digital era, the events industry—from corporate conferences to large-scale festivals—demands the precise synchronization of multiple independent service providers. Managing these logistics manually via fragmented tools is inefficient and prone to error. **Event-Sync** is a centralized web ecosystem designed to bridge the gap between client vision and vendor execution. By providing a unified command center, the platform allows clients to orchestrate complex bookings across diverse categories, while offering vendors an integrated suite for real-time scheduling, contract oversight, and streamlined communication.

II. Statement of the Problem

Traditional event planning is plagued by fragmentation and "information silos." Currently, clients communicate with caterers, photographers, and venue managers through separate, disconnected channels like phone calls or instant messaging. This leads to several critical issues:

- ★ **Scheduling Overlaps:** A lack of a shared calendar leads to double-booking or venue-vendor timing conflicts.
- ★ **Communication Delays:** Changes in event timelines often fail to reach all vendors simultaneously, leading to service failures on the event day.
- ★ **Payment Complexity:** Managing separate deposits and payment deadlines for five or more vendors is prone to human error and financial disputes.
- ★ **No Centralized Timeline:** There is no "single source of truth" where all stakeholders can view the live progress of event preparations.

III. Objectives

General Objective:

To design and implement a real-time web portal that synchronizes event logistics between clients and multiple vendors.

Specific Objectives:

- ★ To develop a Real-Time Timeline using Socket.io to ensure all stakeholders see schedule updates instantly.
- ★ To create a Vendor Inventory & Availability System allowing service providers to manage their own booking calendars dynamically.
- ★ To implement a Secure Payment and Contract Management Module using JWT authentication and encrypted data storage.

- ★ To build an Interactive Client Dashboard for tracking the total budget and status of all booked vendors in one view.

IV. Significance of the Project

This project is significant as it addresses the logistical complexity of the event industry through advanced Software Engineering. For Clients, it reduces the stress of coordination and prevents financial loss due to scheduling errors. For Vendors, it provides a professional platform to manage their business more efficiently.

Project Title: 3 ResearchSphere: A Research Management and Publication Tracking System for Higher Education Institutions

Introduction

ResearchSphere is a centralized, web-based Research Management and Publication Tracking System designed to digitize and streamline the entire research lifecycle—from proposal submission to publication indexing and impact analytics. The system provides structured data storage, automated tracking, real-time dashboards, and role-based access for administrators, faculty members, and department heads.

The platform aims to improve transparency, accountability, and data-driven decision-making within academic institutions.

Statement of the Problem

Research administration in many universities faces significant structural and operational challenges:

- Fragmented Data Silos
- Inaccurate Performance Tracking
- Manual Reporting Burden
- Poor Collaboration Visibility
- Missed Funding Opportunities

Without centralized tracking of grant deadlines and submissions, institutions risk missing funding calls or duplicating efforts.

These inefficiencies reduce institutional competitiveness and create administrative overhead.

Objectives

General Objective

To design and implement a secure, scalable, web-based Research Management and Publication Tracking System that centralizes research data and automates performance monitoring.

Specific Objectives

- To implement a centralized research database that stores projects, publications, grants, and researcher profiles in a normalized relational schema.
- To develop a role-based access control (RBAC) system that differentiates permissions among administrators, faculty members, and department heads.

- To create an automated publication tracking module that records journal articles, conference papers, patents, and books with metadata such as DOI, indexing status, and citation count.
- To build an analytics dashboard that visualizes research productivity using key performance indicators (KPIs) such as:

Significance of the Project

Institutional Impact: The system enhances institutional governance by providing real-time insights into research performance. It supports strategic planning, funding allocation, and ranking improvement.

Administrative Efficiency: By automating reporting and centralizing data storage, administrative workload is significantly reduced. Decision-makers gain immediate access to reliable, structured information.

Faculty Empowerment: Researchers can maintain a verified, organized portfolio of their scholarly output, improving transparency during performance evaluations and promotion reviews.